[Normal] Catalog Article - Request memory dumps from 3 replicas for specific customer application

By João Genebra CURRENT VERSION 1 min 6 Add a reaction

This Change is part of the Backend Runtime Change Catalog.

1. Summary

Apps from our customer, Entel, are currently scaled to the max number of replicas (16) allowed in the ODC infrastructure. The replicas have an average memory usage above the scale down target defined on HPA, even when idle, so the app is not scaling down as expected.

A memory dump was already obtained previously, but more information is necessary.

Two dumps should be collected from replicas with a long up time (19+ hours). If there are no replicas old enough for this app, we should postpone for the next day and re-evaluate.

A third dump has to be collected from a pod that just started. For this we need to kill one of the pods and collect the dump from the new pod that will start up.

For this, we will need to collect 3 memory dumps from replicas of the application.

2. General Information

Change Type	Normal
Change Risk	Low
Downtime Involved	Yes, but should be done out of hours for the customer (between 9am and 12pm UTC+1).

Implementation Team	Backend Runtime
Component(s)/Asset(s)	Application Runtime
Impact Description	App will freeze while memory dump is generated.

3. Implementation Plan

- Collect memory dump from the following application on Entel's producti environment:
  - Stage: Production
  - Cluster: `runp-ga-sa-east-1-01`
  - Namespace: `2bb8945e-d6e5-47d7-9143-b1d82a196eb8`
  - Application Name: Decouple Shopping Cart API
  - Application Key: `fe86f841-f5cf-4d4b-b4c1-76a38b400eea`

To collect the memory dump, follow the **On-demand** section detailed in this document. Disregard the other sections, as they will not be relevant in this scenario.

should already be installed in the directory `/opt/dotnet-tools`. The memory dump should be output to `/tmp` since other directories might require privilege escalation, which is not possible inside app pods. Example for dotnet-dump:

By default, the application runs with PID 1. However, if the application is running under sh (applies to the development environment) you'll need to obtain the dotnet application PID:

```
apt update && apt install -y procps && ps x
```

Collect memory dump

```
/opt/dotnet-dump collect -o /tmp/memory.dmp -p <dotnet proce
```

Copy `memory.dmp` file to the local machine

```
 kubectl cp <pod>:/tmp/memory.dmp memory.dmp
```

4. Rollback plan

N/A

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